

Ordered Radial Trajectories in Acceleration Space Reveal the Internal Structure of the Radial Acceleration Relation

Abstract

The radial acceleration relation (RAR) establishes a remarkably tight empirical correlation between the observed gravitational acceleration in galaxies and that predicted from their baryonic mass distributions. Previous studies have primarily examined the RAR as a population-level relation constructed from measurements spanning many galaxies. However, galaxy rotation curves contain an intrinsic radial ordering that permits investigation of how the relation is realized within individual systems.

Using rotation curve data from the SPARC database, we reconstruct trajectories in acceleration space by connecting successive measurements of baryonic acceleration and observed acceleration in order of increasing galactocentric radius. This approach transforms the conventional point-based representation of the RAR into a set of ordered radial paths that trace the internal dynamical evolution of individual galaxies.

We find that galaxies follow coherent and continuous trajectories in the $(\log g_{\rm bar}, \log g_{\rm obs})$ plane, evolving systematically from high-acceleration inner regions toward low-acceleration outer regions. Despite substantial diversity in morphology, mass, and surface brightness, these trajectories closely follow the mean radial acceleration relation. The observed population-level RAR therefore emerges not merely as a statistical correlation across galaxies, but as the collective expression of ordered radial evolution within individual systems.

These results provide a complementary perspective on the origin and interpretation of the radial acceleration relation. Any successful explanation of galaxy dynamics—whether based on dark matter, modified gravity, or alternative frameworks—must account not only for the existence of the population-level relation but also for its realization as a structured radial trajectory within individual galaxies.

Keywords

Galaxies: Kinematics and dynamics, galaxies: Rotation curves, dark matter, gravitation, galaxies: Structure, methods: Data analysis

1. Introduction

Galaxy rotation curves remain one of the principal observational probes of gravitational dynamics on galactic scales. Measurements extending beyond the optical radii of galaxies have long revealed discrepancies between the observed rotational velocities and those predicted from the distribution of luminous matter alone (Rubin, Ford & Thonnard 1980; Bosma 1981). These discrepancies motivated the development of dark matter halo models within the Λ CDM paradigm and continue to provide important constraints on theories of galaxy formation and evolution.

An important empirical development in this context is the radial acceleration relation (RAR), which links the observed gravitational acceleration, ($g_{\rm obs}$), to the acceleration predicted from the observed baryonic mass distribution, ($g_{\rm bar}$) (McGaugh, Lelli & Schombert 2016). Remarkably, measurements spanning a wide range of galaxy masses, morphologies, and surface brightnesses follow a tight relation with relatively small intrinsic scatter. The existence and robustness of the RAR have motivated extensive theoretical investigation within both dark matter and modified gravity frameworks (Famaey & McGaugh 2012; Ludlow et al. 2017; Keller & Wadsley 2017).

Most analyses of the RAR treat it as a population-level relation. Individual measurements from many galaxies are combined and examined collectively in the $((g_{\rm bar}, g_{\rm obs}))$ plane, revealing a well-defined empirical trend. While this approach has been highly successful in characterizing the relation, it does not explicitly consider the ordered radial structure present within individual galaxy rotation curves.

Rotation curves contain information not only about the magnitude of gravitational acceleration but also about its systematic variation with galactocentric radius. Each galaxy therefore contributes a sequence of measurements that traces a path through acceleration space as one moves from the inner to the outer regions of the system. This radial ordering is typically lost when measurements are considered solely as members of an aggregate population.

The present work investigates how the radial acceleration relation is realized within individual galaxies. Using rotation curve data from the SPARC database (Lelli, McGaugh & Schombert 2016), we reconstruct ordered trajectories in acceleration space by connecting successive measurements according to galactocentric radius. This approach allows the internal structure of the relation to be examined directly.

The central question addressed in this paper is straightforward: does the radial acceleration relation arise solely as a statistical correlation across galaxy populations, or is it also manifested as an ordered radial trajectory within individual systems?

We show that individual galaxies trace coherent and continuous paths in acceleration space that closely follow the mean radial acceleration relation. The results demonstrate that the RAR is not merely a population-level correlation but is also realized through structured radial evolution within galaxies. This finding provides an additional empirical constraint that any successful explanation of galaxy dynamics must reproduce.

2. Data and Definitions

2.1 SPARC Galaxy Sample

This analysis uses data from the SPARC (Spitzer Photometry and Accurate Rotation Curves) database (Lelli, McGaugh & Schombert 2016), which provides high-quality rotation curves and baryonic mass models for 175 nearby disc galaxies spanning a wide range of masses, luminosities, surface brightnesses, and morphological types.

The SPARC sample combines extended H I rotation curves with homogeneous near-infrared photometry at 3.6 μm , enabling construction of detailed baryonic mass distributions and corresponding gravitational acceleration profiles. The dataset has become a widely used benchmark for investigations of galaxy dynamics, the baryonic Tully–Fisher relation, and the radial acceleration relation.

For each galaxy, the database provides measurements of:

- galactocentric radius, (r) ;
- observed circular velocity, $(V_{\text{obs}}(r))$;
- gas contribution, $(V_{\text{gas}}(r))$;
- stellar disc contribution, $(V_{\text{disk}}(r))$;
- bulge contribution, $(V_{\text{bulge}}(r))$, where applicable.

2.2 Baryonic Acceleration

Following standard SPARC analyses, the baryonic circular velocity is computed as

$$V_{\text{bar}}^2(r) = V_{\text{gas}}^2(r) + Y_{\text{disk}} V_{\text{disk}}^2(r) + Y_{\text{bulge}} V_{\text{bulge}}^2(r)$$

where Y_{disk} and Y_{bulge} denote the adopted stellar mass-to-light ratios for the disc and bulge components.

The observed and baryonic gravitational accelerations are then defined by

$$g_{\text{obs}}(r) = V_{\text{obs}}^2(r) / r$$

and

$$g_{\text{bar}}(r) = V_{\text{bar}}^2(r) / r.$$

Throughout this work, accelerations are expressed in cgs units (cm s^{-2}), consistent with previous studies of the radial acceleration relation.

2.3 Acceleration-Space Representation

The radial acceleration relation is commonly represented in logarithmic coordinates,

$$[x = \log_{10}(g_{\rm bar})]$$

$$[y = \log_{10}(g_{\rm obs})]$$

where each measurement contributes a single point in acceleration space.

In conventional analyses, measurements from many galaxies are combined to form a population-level relation. In the present work, however, the radial ordering of measurements within each galaxy is retained. This allows individual galaxies to be represented as ordered sequences of points in the $(\log g_{\rm bar}, \log g_{\rm obs})$ plane and provides a direct means of examining how the radial acceleration relation is realized internally within galaxy rotation curves.

3. Methodology

The objective of this study is to examine how the radial acceleration relation is realized within individual galaxies while preserving the intrinsic radial ordering present in rotation curve measurements. Rather than treating each measurement as an independent point contributing to a population-level correlation, we reconstruct the sequence of accelerations encountered as one moves outward through a galaxy.

3.1 Construction of Acceleration Trajectories

For each galaxy in the SPARC sample, the observed and baryonic accelerations were computed using the definitions given in Section 2. The resulting values were transformed into logarithmic coordinates,

$$[x = \log_{10}(g_{\rm bar})]$$

$$[y = \log_{10}(g_{\rm obs})]$$

The measurements for each galaxy were then ordered according to increasing galactocentric radius. Successive points were connected in this order to generate a continuous trajectory in acceleration space.

This procedure transforms the conventional point-based representation of the radial acceleration relation into a trajectory-based representation in which each galaxy traces a path through the $(\log g_{\rm bar}, \log g_{\rm obs})$ plane. The trajectory therefore preserves information regarding the radial evolution of the system that is not visible when measurements are considered solely as members of an aggregate population.

3.2 Comparison with the Population-Level Relation

To assess the relationship between individual trajectories and the global radial acceleration relation, the reconstructed trajectories were examined relative to the distribution of SPARC measurements in acceleration space.

Particular attention was given to:

- the overall continuity of individual trajectories;
- the direction of radial evolution from inner to outer regions;
- the degree to which trajectories follow the mean RAR trend;
- variations in trajectory morphology among galaxies of different masses and structures.

The analysis is primarily geometric and empirical in nature. No alternative gravitational model, dark matter profile, or modified gravity prescription is fitted to the data. The objective is instead to determine whether the observed population-level relation is reflected in the internal radial organization of individual galaxies.

3.3 Scope of the Analysis

The present study focuses on the structural realization of the radial acceleration relation rather than on the detailed physical mechanisms responsible for it. The analysis therefore does not seek to distinguish between competing theoretical interpretations of the RAR. Instead, it identifies an observational property of the relation that any successful theoretical framework must reproduce.

By retaining the radial ordering inherent in galaxy rotation curves, the method provides a complementary perspective on the radial acceleration relation and its manifestation within individual galactic systems.

4. Results

4.1 Ordered Radial Trajectories in Acceleration Space

The reconstructed trajectories reveal that individual galaxies do not populate the radial acceleration relation (RAR) as collections of independent points. Instead, they trace coherent and continuous paths through acceleration space when measurements are connected in order of increasing galactocentric radius.

For the galaxies examined, trajectories generally originate in the high-acceleration regime associated with the inner regions of the galaxy and progress toward the low-acceleration regime characteristic of the outer disc. Although trajectory shapes vary among galaxies, the overall evolution is highly ordered and exhibits a common directional progression across the sample.

Paper 12 – Figure 1: Galaxy Trajectories in Acceleration Space (SPARC)

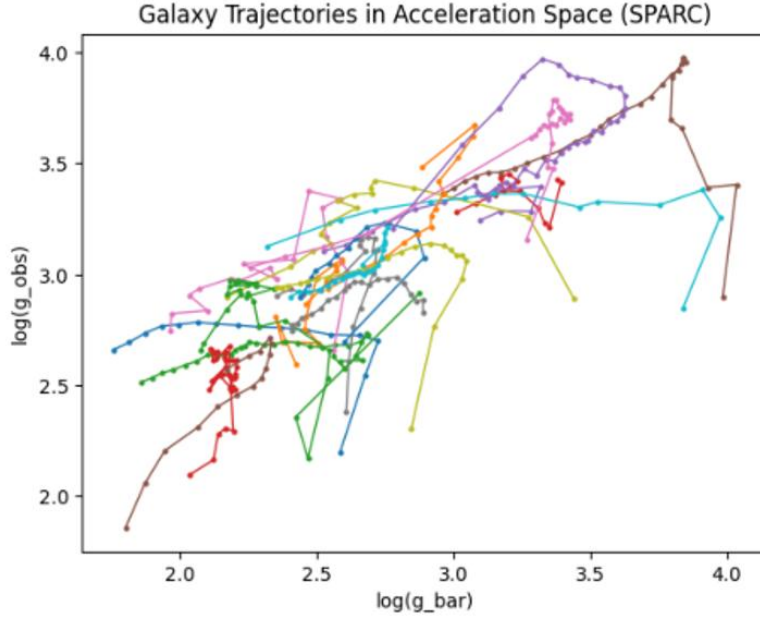


Figure 1. Representative SPARC galaxies reconstructed as ordered trajectories in acceleration space. Successive measurements are connected according to increasing galactocentric radius. Individual systems trace coherent paths from high-acceleration inner regions toward low-acceleration outer regions.

The existence of these ordered trajectories demonstrates that the radial acceleration relation possesses an internal structure that is not immediately apparent in conventional point-based representations.

4.2 Relationship to the Mean Radial Acceleration Relation

To determine whether the observed trajectories are related to the global RAR, the reconstructed paths were examined relative to the population-level distribution of SPARC measurements.

The trajectories closely follow the locus defined by the mean radial acceleration relation. As galaxies evolve through acceleration space with increasing radius, their paths remain largely confined to the same region occupied by the global relation. This behavior indicates that the population-level RAR is not generated by unrelated measurements drawn from different galaxies, but is instead the collective manifestation of ordered radial evolution occurring within individual systems.

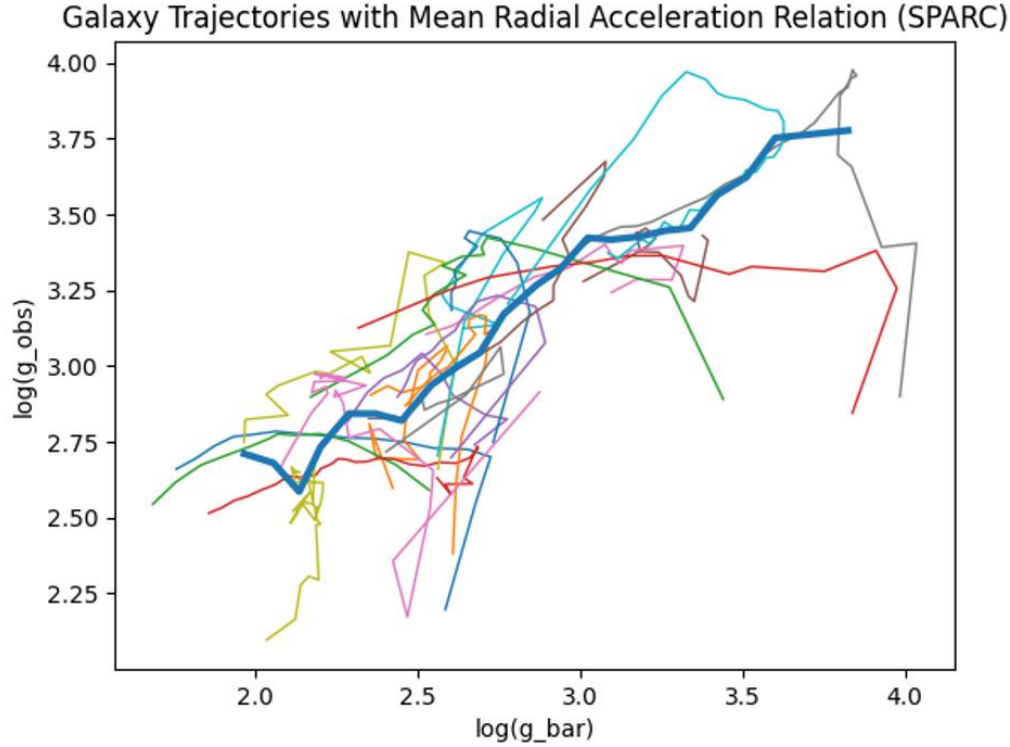


Figure 2. Ordered galaxy trajectories overlaid on the population-level radial acceleration relation. Individual trajectories closely follow the empirical relation, demonstrating that the RAR is realized internally within galaxies as well as across the galaxy population.

While individual systems exhibit modest deviations from the mean trend, the overall alignment remains striking across a broad range of galaxy masses and morphologies.

4.3 Structural Realization of the RAR

The principal result of this study is that the radial acceleration relation is realized internally within galaxies as a structured radial sequence.

Conventionally, the RAR is interpreted as a tight empirical relation connecting measurements obtained from many galaxies. The present analysis demonstrates that the same relation emerges naturally within individual galaxies when their radial ordering is preserved. Each galaxy contributes not merely a collection of points to the RAR, but an ordered trajectory that traverses the relation in a systematic manner.

This observation provides an additional empirical property of the RAR that must be explained by successful models of galaxy dynamics. Any theoretical framework capable of reproducing the population-level relation must also account for its realization as a continuous radial trajectory within individual galaxies.

4.4 Robustness of the Result

The trajectory structure arises directly from the observed radial ordering of SPARC rotation-curve measurements and does not depend on additional fitting procedures, parameter optimization, or model assumptions beyond the standard construction of baryonic and observed accelerations.

The observed behavior is therefore a geometric property of the data itself. While the physical interpretation of the radial acceleration relation remains an open question, the existence of ordered radial trajectories constitutes an empirical feature that is independent of whether the underlying explanation is attributed to dark matter, modified gravity, or another mechanism. The results presented here demonstrate that the radial acceleration relation possesses both a population-level manifestation and an internal realization within individual galaxies.

5. Discussion

5.1 The Radial Acceleration Relation as an Internal Dynamical Sequence

The radial acceleration relation is typically presented as a population-level empirical correlation obtained by combining measurements from many galaxies. The results presented here demonstrate that the relation also possesses a clear internal realization within individual systems. When radial ordering is preserved, galaxies trace coherent trajectories that closely follow the same acceleration-space locus defined by the global RAR.

This finding suggests that the RAR is not merely an emergent statistical property of galaxy populations. Rather, it reflects an ordered dynamical sequence through which individual galaxies evolve as a function of galactocentric radius. The population-level relation can therefore be viewed as the aggregate manifestation of a process already present within individual systems.

5.2 Implications for Galaxy Dynamics

The present analysis does not favor a particular theoretical explanation for the radial acceleration relation. However, it introduces an additional empirical constraint that successful models must satisfy. Within modified gravity frameworks such as MOND, the RAR emerges as a deterministic relationship between baryonic and observed accelerations. The existence of ordered radial trajectories is therefore broadly consistent with the expectation that acceleration should evolve continuously along a well-defined relation.

Within Λ CDM, the RAR is generally understood as an emergent consequence of the interaction between baryonic matter, dark matter halos, and galaxy formation processes. Numerical simulations have demonstrated that relations resembling the observed RAR can arise naturally under appropriate conditions. The present results imply that such models must reproduce not only the global relation itself but also the ordered radial trajectories through which individual galaxies populate that relation.

Regardless of the underlying mechanism, the trajectory structure identified here represents an observational property that any successful model of galaxy dynamics should be capable of explaining.

5.3 Relation to Previous Work

Previous studies have established the existence, tightness, and universality of the radial acceleration relation across a broad range of galaxy types (McGaugh et al. 2016; Lelli et al. 2016; Li et al. 2018). Additional investigations have examined scatter about the relation, its connection to the baryonic Tully–Fisher relation, and its implications for dark matter and modified gravity models.

The present work differs in emphasis. Rather than focusing on the statistical properties of the RAR itself, it examines how the relation is populated internally within galaxies. By retaining the radial ordering inherent in rotation-curve measurements, the analysis reveals a geometric structure that is largely hidden when measurements are treated solely as members of an aggregate population. This perspective does not replace existing analyses of the RAR. Instead, it complements them by highlighting an additional empirical feature of the data that may provide useful constraints for future theoretical and numerical studies.

5.4 Limitations and Future Work

The present study is intentionally descriptive and focuses on the geometric structure of galaxy trajectories in acceleration space. No attempt has been made to quantify trajectory morphology, classify trajectory families, or compare trajectories directly with predictions from specific theoretical models.

Several extensions suggest themselves naturally. Future work could investigate systematic variations in trajectory shape as a function of galaxy mass, surface brightness, morphology, or environment. Comparisons with hydrodynamical simulations and modified gravity predictions may also help determine whether different theoretical frameworks generate distinguishable trajectory signatures. More generally, the trajectory-based representation introduced here provides a new way of examining the internal realization of empirical scaling relations and may prove useful in the study of other dynamical correlations in galaxy evolution.

6. Conclusions

Using rotation-curve data from the SPARC database, we have investigated how the radial acceleration relation (RAR) is realized within individual galaxies when the intrinsic radial ordering of measurements is preserved.

By reconstructing trajectories in acceleration space through sequential connection of measurements in order of increasing galactocentric radius, we find that galaxies trace coherent and continuous paths in the $(\log g_{\rm bar}, \log g_{\rm obs})$ plane. These trajectories evolve from the high-acceleration inner regions of galaxies toward the low-acceleration outer regions and closely follow the locus defined by the population-level radial acceleration relation.

The principal results of this study may be summarized as follows:

1. Individual galaxies trace ordered trajectories in acceleration space rather than contributing only isolated measurements to the RAR.
2. These trajectories closely follow the empirical radial acceleration relation across a broad range of galaxy properties.
3. The population-level RAR emerges not only as a correlation among measurements drawn from many galaxies, but also as a structured radial sequence realized within individual systems.
4. The ordered trajectory structure represents an additional observational property that successful models of galaxy dynamics should reproduce.

The analysis presented here is intentionally model-independent. No assumptions are made regarding whether the radial acceleration relation arises from dark matter, modified gravity, baryonic feedback processes, or another physical mechanism. Instead, the study identifies an empirical feature of the data itself.

The results suggest that preserving radial ordering provides a complementary perspective on the radial acceleration relation and reveals an internal structure that is largely obscured in conventional point-based representations. Future investigations comparing observed trajectories with theoretical and numerical predictions may help determine whether different models of galaxy dynamics produce distinct acceleration-space signatures. More broadly, the trajectory-based approach introduced here illustrates how information contained in the ordered structure of observational data can provide additional constraints beyond those available from population-level correlations alone.

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Data Availability

The data underlying this article are publicly available through the SPARC (Spitzer Photometry and Accurate Rotation Curves) database described by Lelli, McGaugh & Schombert (2016).

SPARC data may be accessed at:

<http://astroweb.cwru.edu/SPARC/>

No new observational data were generated for this study. All results were obtained through analysis of publicly available SPARC rotation-curve measurements and derived baryonic mass models.

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Barry Kieran Radcliffe conceived the study, performed the data analysis, interpreted the results, prepared the figures, and wrote the manuscript.

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Conflicts of Interest

The author declares no conflicts of interest.

Artificial Intelligence Disclosure

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